



Help Me — Calculation Guide for CAPEX, OPEX & Fiscal Inputs

This guide explains how to populate every advanced CAPEX, OPEX and fiscal field on the Asset Pulse Scenario form (and in the multi-row CSV template at `examples/asset_pulse_scenario_input_template.csv`). Each section gives you the **meaning, how to estimate it, a working formula or example, QA checks, and operator-specific notes** for the three archetypes Asset Pulse is most often used for:

- **NOC** — National Oil Company running an internal screening view (e.g.

Saudi Aramco, ADNOC, Petrobras, Pemex, Sonatrach).

- **IOC** — International Oil Company under a host-government contract such as

a PSC, TSC or concession (e.g. ExxonMobil, Shell, TotalEnergies, BP).

- **US Independent** — onshore US working-interest operator under

lease-royalty + severance-tax economics (e.g. shale operator in the Permian, Eagle Ford, Bakken).

> **Disclaimer:** Asset Pulse is a decision-support tool. The numbers below > are typical reference ranges drawn from the public sources cited inline. > They are not legal, tax or accounting advice. Always validate fiscal terms, > royalty schedules and tax rates against the actual contract or local > regulator.

1. Operating cost inputs

1.1 `fixed_opex_usd_month`

Meaning. The lease/facility/operator fixed cost that you incur every month even if the well chokes back to zero — pumper/operator labour, surface equipment rental, SCADA/comms, road and lease maintenance, regulatory reporting. This is the classic "fixed lifting cost" component of LOE.

How to estimate.

1. Pull last 12 months of LOE/AFE actuals from the accounting system.
2. Strip out anything that scales with production (chemicals, electricity,

water disposal, hauling) — those go into the variable buckets below.

3. Divide the remaining fixed annualised total by 12.

Formula.

`fixed_opex_usd_month` = (annual fixed lease + facility + labour cost) / 12

Typical ranges (per the [EIA upstream cost study](#) and [Lumel LOE breakdown](#)):

Asset class	Fixed OPEX USD/month
US shale single horizontal well	8,000 – 25,000
Conventional onshore stripper	1,500 – 6,000
Mature giant ME field (per well)	4,000 – 15,000



Asset class	Fixed OPEX USD/month
Offshore platform (per well slot)	50,000 – 300,000

QA checks.

- Re-derive on a \$/BOE basis and compare with the [EIA performance profile lifting-cost band](#).
- Make sure you have **not** double-counted variable opex.
- Pumper labour should appear here, not in `variable_opex_usd_boe`.

Operator notes.

- **NOC**: Use internal LOE ledger; many giant fields run very low fixed

per-well costs because headcount and surface infrastructure are pooled across the field. State subsidies on diesel/power should be reflected at the *delivered* (post-subsidy) price the operator actually pays.

- **IOC**: Use the contractor-bookable share of operating cost, as defined in

the cost-recovery procedures of the PSC/TSC. Some contracts cap technical assistance fees, IT/G&A allocations and parent-company charge-backs.

- **US Independent**: Equals the fixed component of LOE per well. If the

lease has multiple wells, allocate the lease-level fixed cost across active wells using a defensible method (equal split, BOE-weighted, or working interest weighted).

1.2 `variable_opex_usd_boe`

Meaning. All operating cost that scales with production volume — chemicals, electricity for artificial lift, compression fuel, hauling, chemical inhibitors, downhole intervention. Charged on every BOE produced.

How to estimate.

1. Sum last 12 months of variable LOE line items.
2. Divide by total BOE produced over the same period.

Formula.

`variable_opex_usd_boe` = annual variable LOE / annual produced BOE

Typical ranges (cross-checking [Lumel's LOE article](#) against [EIA finance performance profiles](#)):

Asset class	Variable OPEX USD/BOE
US shale, low water cut	2.5 – 6
US shale, mature high water cut	6 – 15
ME giant onshore, very low water cut	1 – 3
Heavy oil / thermal recovery	8 – 25

QA checks.

- Don't include water-handling cost here — it has its own field.



- Avoid double-counting electricity if your power is netted out of revenue.

Operator notes.

- **NOC:** Use realised internal \$/BOE; remember that subsidised utilities

understate cost relative to international benchmarks.

- **IOC:** Make sure variable costs are *recoverable* under the cost-recovery

rules. Some PSCs disallow specific items (entertainment, parent G&A above a cap, financing cost).

- **US Independent:** Pump electricity, gas-lift fuel and methanol injection

are the typical drivers. Many independents compute this directly from their LOS (Lease Operating Statement).

1.3 water_handling_usd_bbl

Meaning. The marginal cost of dealing with one barrel of produced water — gathering, lift, treatment, hauling, salt-water disposal (SWD) injection fees. Quoted per **barrel of water**, not per BOE.

How to estimate.

$$\text{water_handling_usd_bbl} = (\text{annual water OPEX} + \text{SWD} / \text{disposal fees}) / \text{annual water bbl}$$

Reference the water cost discussion in [the Lumel LOE article](#) and the EIA's note that water handling is a major variable LOE driver in mature wells ([EIA upstream costs](#)).

Typical ranges.

System	USD/bbl water
Pipeline-connected SWD, dry area	0.25 – 0.75
Trucked SWD	1.50 – 4.00
Offshore overboard treatment	0.10 – 0.30
Steam/thermal cycle reuse	0.50 – 2.50

QA checks.

- Pair with water_cut_pct — if water cut grows in the projection, the

implied monthly water cost grows with it. Check that the resulting \$/BOE water spend in late-life months is still credible.

- Don't include water *injection* CAPEX here — that goes into capex_total_usd.

Operator notes.

- **NOC:** Brownfield giant fields often have purpose-built water injection

and disposal grids; per-barrel cost can be very low (< \$0.30) but trends up as water cut climbs above 85%.

- **IOC:** Confirm whether produced water handling is a recoverable cost or

treated as the contractor's risk inside the operating fee.

- **US Independent:** Often the single largest variable cost in mature



Permian wells. Trucking adds \$1–\$3/bbl over pipeline; line losses are another 2–5%.

1.4 water_cut_pct

Meaning. Water as a **percentage of total liquid production** at the end of the projection horizon (or starting water cut, depending on which UI field you are filling). Asset Pulse linearly interpolates between start and end water cut over the horizon.

How to estimate. Read the most recent month's well test or production allocation; project forward using analog wells in the same reservoir.

Formula.

$$\text{water_cut_pct} = 100 * \text{water_bbl} / (\text{water_bbl} + \text{oil_bbl})$$

Typical ranges.

- New shale well, year 1: 5 – 30%
- US shale, mature: 60 – 90%
- ME giant onshore mature: 30 – 60% (peripheral water injection)
- North Sea late-life: 80 – 95%

QA checks.

- Enter as a **percent** in CSV (e.g. 45 for 45%); the importer converts to a fraction internally.
- End-of-life water cut > 95% is rare — sanity check before locking in.

Operator notes.

- **NOC:** For peripheral or pattern waterflood, end water cut is often a reservoir-engineering output, not a free input.
- **IOC:** A higher water cut in a PSC reduces produced oil and therefore shrinks both the cost-recovery base (cost oil) and the contractor's profit oil; it materially hurts contractor NPV.
- **US Independent:** Strongly affects netback because every incremental water barrel costs water_handling_usd_bbl on top of pump electricity.

2. Scenario multipliers

2.1 capex_multiplier

Meaning. A **ratio** applied to *all* CAPEX (development + sustaining). 1.0 means use base CAPEX as entered. 1.25 means assume a 25% capital overrun. 0.9 means a 10% saving versus the base AFE.

How to set.

- Use 1.0 for the deterministic base case.



- Use historical AFE-vs-actual variance to centre Monte Carlo distributions.

Many basins show a P90 overrun of 15–35% on offshore and 5–15% on development drilling.

Formula.

`adjusted_capex = base_capex * capex_multiplier`

Operator notes.

- **NOC:** Frequently used for fiscal stress tests against

state-budget-driven price decks.

- **IOC:** The CAPEX multiplier interacts with `capex_uplift_pct` under

PSCs — overrun hits cash flow but uplift only applies to the *base* recoverable CAPEX as defined by the contract.

- **US Independent:** Tornado/Monte Carlo this between 0.9 and 1.3 to capture

service-cost inflation and rig availability risk.

2.2 `opex_multiplier`

Meaning. Same idea as the CAPEX multiplier, applied to all OPEX (fixed + variable + water handling). 1.0 is base. 1.1 is 10% inflation.

How to set. Use service-cost inflation indices (BLS for US, IEA for international); 1.05–1.15 is typical for stress cases.

Operator notes.

- **NOC:** Often used to model future labour cost increases under local

content rules.

- **IOC:** An `opex_multiplier > 1.0` on a TSC chips into the recoverable

cap directly because TSCs reimburse cost up to a cap.

- **US Independent:** Critical lever — LOE inflation has historically been

the second-biggest driver of breakeven oil price after the price deck itself.

2.3 `downtime_months` and `downtime_start_month`

Meaning. A contiguous block of months with zero production (e.g. turnaround, pipeline outage, weather shut-in). `downtime_start_month` is 1-indexed.

How to set.

- For unscheduled risk events, use Monte Carlo with a Poisson-style

occurrence and 1–3 month duration.

- For a planned turnaround, set `downtime_months` and `downtime_start_month`

directly and re-run the scenario.

Formula behaviour. During the downtime block:



- Oil, gas, water volumes go to 0.
- Variable OPEX and water handling drop to 0.
- Fixed OPEX continues (the asset is still being kept hot).
- Royalty, severance and PSC profit oil also drop with revenue.

QA checks.

- `downtime_start_month + downtime_months ≤ horizon_months` — otherwise

most of the downtime falls beyond the projection.

- Set both to 0 to disable.

Operator notes.

- **NOC**: Use to model major maintenance turnarounds in long-life giant

fields.

- **IOC**: PSC cost recovery is limited to the cost-oil **ceiling × revenue**

— when revenue drops to zero in a downtime month, no costs are recoverable that month. Costs carry forward.

- **US Independent**: Use to test winter freeze-offs (Bakken/Anadarko) or

permit-driven shut-ins.

3. PSC fiscal inputs (`psc_cost_recovery` regime)

A production sharing contract (PSC) splits production between host government and contractor through royalty → cost recovery → profit oil → contractor tax. The mechanics below match the Asset Pulse PSC engine and the descriptions in [the PetroSkills introduction to PSCs](#), the [Bindemann \(1999\) Oxford Energy paper](#), and the [Oklahoma Law Review summary of PSC structures](#).

3.1 `psc_royalty_rate_pct`

Meaning. A royalty taken off the top of gross petroleum revenue **before** cost recovery. Some modern PSCs have no royalty (set to 0).

Typical ranges. 0% – 15% (common: 10%). Bindemann (1999) notes royalty is often 8–15% where present.

Formula.

```
royalty_$ = gross_revenue * psc_royalty_rate_pct / 100
post_royalty_revenue = gross_revenue - royalty_$
```

QA checks. Enter percent (10 for 10%), not fraction.

Operator notes.

- **NOC**: Often run with royalty = 0 internally, since royalty is just a

bookkeeping transfer between state and state-owned operator.

- **IOC**: Royalty reduces both the cost-recovery base **and** the profit



oil base — it is the single most punitive PSC term for a low-margin field.

3.2 psc_cost_recovery_ceiling_pct

Meaning. The **maximum percentage** of (post-royalty) revenue the contractor can use to recover petroleum costs in any given period. Unrecovered costs carry forward. Typical: 50%–80% (e.g. Indonesia historic 80% cost-oil ceiling).

Formula.

available_cost_oil	=	post_royalty_revenue * psc_cost_recovery_ceiling_pct / 100
recoverable_pool	=	carry_forward + opex + capex * (1 + capex_uplift_pct/100)
recovered_\$	=	min(recoverable_pool, available_cost_oil)
new_carry_forward	=	recoverable_pool - recovered_\$

QA checks. A high ceiling (>80%) speeds up cost recovery but can run afoul of profit-oil minima in some contracts. Make sure your ceiling is consistent with the contract.

3.3 psc_contractor_profit_oil_share_pct

Meaning. The contractor's **share of profit oil** after cost recovery. Profit oil = post-royalty revenue minus recovered cost. Typical: 20%–50%, often on a sliding scale by R-factor or cumulative production.

Formula.

profit_oil	=	post_royalty_revenue - recovered_\$
contractor_profit_oil	=	profit_oil * psc_contractor_profit_oil_share_pct / 100
gov_profit_oil	=	profit_oil - contractor_profit_oil

QA checks. Asset Pulse uses a single flat contractor share — if the contract has an R-factor scale, pick a representative average for the period or run multiple scenarios for the lower/upper plateau.

3.4 contractor_tax_rate_pct

Meaning. The income tax the contractor pays on its taxable petroleum income (recovered cost + contractor profit oil, depending on the regime). In many PSCs cost recovery is *not* taxable and only profit oil is — Asset Pulse applies tax on contractor profit oil + remuneration (TSC) only.

Typical ranges. 25% – 55%. Many ME PSCs land around 30–50%.

Formula.

contractor_after_tax = recovered_\$ + contractor_profit_oil * (1 - contractor_tax_rate_pct/100)

Operator notes. This field is reused by the TSC engine to tax remuneration income.

3.5 capex_uplift_pct

Meaning. Many PSCs allow contractors to recover **more than nominal CAPEX** (e.g. 110%, 120%) as compensation for time-value-of-money on capital locked into cost recovery. 0 = no uplift.

Typical ranges. 0% – 25%.

Formula — already shown in §3.2 (capex * (1 + capex_uplift_pct/100)).

QA checks. Don't double-count: if you already grossed up base CAPEX manually, leave uplift at 0%.



4. TSC fiscal inputs ([technical_service_contract](#) regime)

Technical service contracts (Iraq-style) reimburse the contractor for recoverable cost up to a **cap** plus a flat **remuneration fee per BOE**. The contractor never owns hydrocarbons. Source: [2B1st Consulting on TSCs](#) and the [Cornell service contracts review paper](#).

4.1 `tsc_fee_usd_boe`

Meaning. The flat per-BOE remuneration the contractor earns, often \$1–\$3/BOE for Iraq-class TSCs.

Formula.

`remuneration_$` = `produced_BOE` * `tsc_fee_usd_boe`

QA checks. Iraq tender bids in 2009/2010 ranged roughly \$1.15 – \$5.50/BOE; above \$5/BOE is unusual today.

4.2 `tsc_cost_recovery_cap_usd_month`

Meaning. A **monthly cap** on petroleum cost reimbursement to the contractor (in absolute USD/month, not as % of revenue). Eligible costs above the cap roll forward to future months.

Formula.

<code>recoverable_pool</code>	=	<code>carry_forward</code> + <code>opex</code> + <code>capex</code> * (1 + <code>capex_uplift_pct</code> /100)
<code>recovered_\$</code>	=	<code>min</code> (<code>recoverable_pool</code> , <code>tsc_cost_recovery_cap_usd_month</code>)
<code>new_carry_forward</code>	=	<code>recoverable_pool</code> - <code>recovered_\$</code>
<code>contractor_cash_\$</code>	=	(<code>recovered_\$</code> + <code>remuneration_\$</code>) * (1 - <code>contractor_tax_rate_pct</code> /100)

QA checks. Set the cap to a realistic absolute number — e.g. \$3.5M/month on a 180 kBOPD field is a tight 0.6 \$/BOE recovery limit.

Operator notes.

- **IOC:** A TSC behaves more like a fee-for-service contract than a PSC.

The contractor's economics are driven almost entirely by the fee and the cap — production upside accrues to the host.

- **NOC:** TSC is the typical **inverse** view: the host keeps essentially

all profit oil and pays a defined fee.

5. Concession fiscal inputs ([concession_tax_royalty](#) regime)

Concession (or tax-and-royalty) regimes assign hydrocarbons to the licensee who pays royalty on gross revenue and corporate income tax on taxable profit. Source: [ScienceDirect on tax-royalty regimes](#), the [petroleum fiscal regime overview](#), and the [Meehan LinkedIn primer on concession regimes](#).

5.1 `concession_royalty_rate_pct`

Meaning. Royalty taken off gross revenue. Asset Pulse also supports a Saudi-style **progressive** royalty: 15% on the first \$70/bbl, 45% from \$70–\$100, 80% above \$100. Toggle that with `concession_royalty_progressive` in the form / CSV; otherwise the flat rate is used.

Typical ranges. 5% – 20% for flat schedules.

Formula (flat).



$\text{royalty_\$} = \text{gross_revenue} * \text{concession_royalty_rate_pct} / 100$

5.2 concession_income_tax_rate_pct

Meaning. Upstream corporate income tax on taxable profit (revenue – royalty – opex – allowances). Asset Pulse applies it to the post-royalty operating profit. Saudi AGSI-style upstream tax = 50%.

Formula.

$\text{taxable_income} = \text{gross_revenue} - \text{royalty_\$} - \text{opex} - \text{depreciation/CAPEX_amortisation}$
 $\text{income_tax_\$} = \max(\text{taxable_income}, 0) * \text{concession_income_tax_rate_pct} / 100$
 $\text{contractor_cash_\$} = \text{taxable_income} - \text{income_tax_\$}$

QA checks. Confirm tax rate is the **upstream** rate, which is often higher than the general corporate rate (50% AGSI vs 20% general in KSA).

6. NOC vs IOC vs US Independent — what to actually input

Topic	NOC (internal)	IOC (PSC / TSC / concession)	US Independent (royalty / tax)
Fiscal regime	noc_internal	psc_cost_recovery / technical_service_contract / concession_tax_royalty	us_royalty_tax
Pricing	Internal transfer or government-set	Market basis (Brent, ICE), with marketing fee	WTI - basis differential
Royalty / govt take	Often modeled as 0 internally; gov capture via transfer pricing or dividends	Contract royalty + cost recovery + profit oil split	12.5% – 25% lease royalty
Tax	Optional corporate tax (gov_transfer field)	contractor_tax_rate_pct	Severance + ad valorem (DW Energy on severance, Investopedia: severance tax). TX: oil 4.6%, gas 7.5%
OPEX	Subsidised utilities, low fixed/BOE in giant fields	Use <i>recoverable</i> OPEX subset	Full LOE per LOS
CAPEX uplift	Not typical	0% – 25%	Not applicable
Carry-forward of unrecovered cost	Not used	Yes — built into PSC/TSC engines	Not applicable
Working interest / NRI	100% / 100% (state)	Per JOA	WI < 100%, NRI = WI × (1 - royalty - ORRI)



Topic	NOC (internal)	IOC (PSC / TSC / concession)	US Independent (royalty / tax)
Strategic levers	Production continuity, local content, job creation	Cost recovery speed, profit oil share, R-factor	Drilling cadence, hedging, severance optimisation, depletion deductions (Instead on royalty taxation)

> **US tax note.** Severance/production taxes are imposed by individual > states and vary widely. Income-tax effects (depletion, IDC) are > jurisdiction-specific — Asset Pulse does **not** do US tax planning. For an > overview see > [Investopedia on severance tax](#); > for state-by-state rates see > [DW Energy's severance tax overview](#).

7. Worked examples

7.1 US Independent — Permian shale horizontal

```

fiscal_regime           = us_royalty_tax
oil_price_usd_bbl       = 72
initial_rate_boe_month  = 21,000
decline_rate_annual_pct = 45      # entered as percent
months                 = 120
working_interest_pct    = 100
net_revenue_interest_pct = 81.25  # 100 - 18.75 royalty
royalty_rate_pct        = 18.75
severance_tax_rate_pct  = 4.5     # TX oil severance
discount_rate_pct       = 10
capex_total_usd         = 9,290,000
fixed_opex_usd_month    = 14,000
variable_opex_usd_boe   = 4.50
water_handling_usd_bbl  = 1.50
water_cut_pct           = 25     # end-of-horizon
capex_multiplier        = 1.0
opex_multiplier         = 1.0

```

Sanity: total LOE/BOE in year 1 $\approx$ fixed_opex_usd_month / monthly_BOE + variable_opex_usd_boe + water_cost_per_BOE $\approx$ \$0.7 + \$4.5 + (\$1.5 $\times$ 0.33 bbl water/BOE) $\approx$ **\$5.7/BOE**, in line with Permian peers per [EIA performance profiles](#).

7.2 IOC PSC — Indonesia-style cost recovery

```

fiscal_regime           = psc_cost_recovery
oil_price_usd_bbl       = 80
initial_rate_boe_month  = 32,000
decline_rate_annual_pct = 18
months                 = 150
discount_rate_pct       = 10
capex_total_usd         = 42,000,000
fixed_opex_usd_month    = 75,000
variable_opex_usd_boe   = 5.0
water_handling_usd_bbl  = 2.0
water_cut_pct           = 30
psc_royalty_rate_pct    = 10
psc_cost_recovery_ceiling_pct = 80
psc_contractor_profit_oil_share_pct = 40

```



```
contractor_tax_rate_pct      = 30
capex_uplift_pct            = 0
capex_multiplier            = 1.05    # 5% AFE risk
```

Mechanics for month 1 (rough): gross revenue \$76.8M (32 kBOPD × 30 × \$80) → royalty \$7.68M → post-royalty \$69.1M → cost-oil ceiling 80% = \$55.3M available → recover ~\$0.85M opex + \$42M capex = \$42.85M, leaving \$26.3M as profit oil → contractor 40% = \$10.5M → tax 30% on profit oil (~\$7.4M after-tax). Contractor also retains the \$42.85M of recovered cost. See the mechanics in the [PetroSkills PSC primer](#).

7.3 NOC — Saudi-style internal screening

```
fiscal_regime                = noc_internal
oil_price_usd_bbl            = 75          # internal flat deck
initial_rate_boe_month       = 45,000
decline_rate_annual_pct      = 12
months                       = 180
discount_rate_pct            = 8           # state cost of capital
capex_total_usd              = 55,000,000
fixed_opex_usd_month         = 90,000
variable_opex_usd_boe        = 3.20
water_handling_usd_bbl       = 0.60       # pipelined SWD
water_cut_pct                = 12
gov_transfer_pct             = 0           # NOC keeps gross
contractor_tax_rate_pct      = 0
```

Use this view for **internal capital allocation, well-prioritisation, and abandonment timing**. Strategic value of long-life production, local content and government employment is *not* in NPV — capture it qualitatively in the Decision Matrix view (see docs/application-help.pplx.md).

8. CSV input QA checklist

Before clicking **Save & Run** in the Scenario tab's Upload data sub-panel:

- [] All *_pct columns entered as **percent** (e.g. 45 for 45%). The

importer converts to fractions automatically.

- [] fiscal_regime is one of us_royalty_tax, noc_internal,

psc_cost_recovery, technical_service_contract, concession_tax_royalty.

- [] PSC rows have psc_royalty_rate_pct,

psc_cost_recovery_ceiling_pct, psc_contractor_profit_oil_share_pct, contractor_tax_rate_pct populated.

- [] TSC rows have tsc_fee_usd_boe, tsc_cost_recovery_cap_usd_month

and contractor_tax_rate_pct.

- [] Concession rows have concession_royalty_rate_pct and

concession_income_tax_rate_pct.

- [] downtime_start_month + downtime_months ≤ months.



- `[ ] capex_multiplier` and `opex_multiplier` set to 1.0 for the base

case unless explicitly modelling overrun.

- `[ ] water_cut_pct` is plausible at horizon end (rarely > 95%).
- `[ ]` No commodity-tax double counting (US severance is folded into

`us_royalty_tax` — don't also re-deduct it in the OPEX line).

9. Where to look next

- App overview and CSV workflow: `[docs/application-help.pplx.md](application-help.pplx.md)`.
- Field-by-field methodology with Middle East context:

`[docs/middle-east-capex-opex-fiscal-methodology.pplx.md](middle-east-capex-opex-fiscal-methodology.pplx.md)`.

- Onshore well CAPEX/OPEX taxonomy:

`[docs/oil-well-capex-opex-knowledge-base.pplx.md](oil-well-capex-opex-knowledge-base.pplx.md)`.

- Multi-regime CSV template: `examples/asset_pulse_scenario_input_template.csv`.